

Traumatic Arrest Guideline

Key Concepts

- **M.A.R. of MARCHES** should be done **before** chest compressions in traumatic arrest
- **Penetrating traumatic arrest** with or without any signs of life –**M.A.R.** and load patient for transport to level 1
- **Blunt traumatic arrest** – perform **M.A.R.** ON SCENE and call Medical Control Physician
- Transport rapidly, ideal scene time < 5 min
- Closed Chest Compressions/LUCAS are of limited value in traumatic arrest and should be **de-emphasized**. Do not delay procedures focused on correcting the traumatic mechanism leading to arrest
- Emphasize blood administration over chest compressions in hemorrhagic shock if available- need to "fill the tank"
- Application of the LUCAS is to be initiated after **M.A.R** only when beginning patient transport
- Field termination not allowed without calling Medical Control Physician unless Dead on Scene criteria is met (see page 16 of General Patient Care)

MARCHES Protocol

- **Massive Hemorrhage** – Bleeding control
- **Airway** – NPA/OPA/SGA/ETT/Cric
- **Respiration**– Decompress chest for tension pneumothorax, vented occlusive dressing for open pneumothorax
- **Circulation** – IV/IO TXA, pelvic binder, wound packing, Whole Blood if appropriate
- **Hypothermia** care
- **Eye injuries** – Cover with rigid shield and no pressure on the eye
- **Spinal motion restriction** if indicated

Signs of Life

- Breathing, even agonal respirations
- PEA
- End Tidal CO2 over 20 after airway has been established and ventilations for one minute
- Corneal Reflex
- Pupillary response to light
- Movement of extremities
- Cardiac motion on Point of Care Ultrasound (US) if available
- Witnessed or confirmed arrest < 5 minutes

EMT/AEMT

- Perform MARCHES according to Trauma General Patient Care Guidelines
- Do not start chest compressions until
 - **Massive bleeding** is controlled, Tourniquets for extremity, Hemostatic packing
 - **Airway** management with BVM / NPA / OPA or SGA as needed
 - **Respirations** are addressed for possible tension hemo/pneumothorax with a minimum of needle decompression bilaterally
- If massive bleeding below the diaphragm suspected – put the IO in the humerus
- Manual chest compressions after the above(LUCAS only to be used in transport)
- **Hypothermia** – minimize heat loss
- Pausing chest compressions to reassess and redo the above is appropriate

EMT/AEMT Expanded Scope

- **M.A.R.** on scene and call for immediately transport if ROSC
 - **Massive Hemorrhage** – Control massive bleeding if not done.
 - **Airway** – Advanced airway management SGA/ETT or surgical cricothyrotomy
 - **Respiration** – Needle Thoracostomy
- **Circulation** – Reassess all wounds for hemostasis. Once decision to transport then:
 - **TXA** 2 grams IV/IO push per TXA COG
 - Pelvic binder for pelvic instability or for blunt mechanism likely to result in pelvic fracture
 - Chest Compressions if pulseless and **M.A.R.** is addressed
- **Hypothermia** – minimize heat loss
- Pausing chest compressions to reassess and redo the above is appropriate

Paramedic

- **See above EMT/AEMT Expanded Scope**
- Consider field termination if worked on scene and M.A.R. has been dealt with
 - Medical Control Physician may request transport based on specifics of the case
 - End Tidal CO2, monitor, physical signs of life, and downtime are factors influencing transport decisions
 - **All patients must be extricated and assessed in a supine position before termination of resuscitation can be considered**

Termination of resuscitation can always be considered in an austere environment when the below conditions are present:

- **Resources** are exhausted
- The scene becomes unsafe for medical and rescue personnel